



# MANDANA KIA

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Magdeburg, Germany

Medical engineer and researcher working at the intersection of healthcare, medical imaging, and robotics. I have over five years of experience in medical engineering, medical image processing, data analysis, and medical affairs. I completed my Master's in Medical Systems Engineering with research in fluoroscopic CT and needle-induced artifact correction. I am currently working as a research associate on AI-based robotic assistance, focusing on humanoid robotics, computer vision, sensor data, and intelligent robotic systems for medical applications.

## EDUCATION

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### **MSc.: Otto- von Guericke University of Magdeburg – Grade: 2.2**

Focus: Medical visualization (Human Computer interfaces in medicine, 3D & advanced interaction, Computer assisted surgery and Visual analytics), Data analysis, Medical/ Hybrid imaging and Research.

2021 – July 2026

Magdeburg, Germany

### **BSc.: Medical University of Tehran, Iran – Grade equivalent: 1.5**

Focus: Medical imaging, IT and Research

Bachelor thesis: Image and data processing – EIT device (Dräger)

2014 – 2018

Tehran, Iran

## WORK EXPERIENCE

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### **Researcher – (Otto- von- Guericke University of Magdeburg & University Clinic Magdeburg: Department of Radiology and Nuclear medicine)**

“AI-Based Humanoid Robotics for Medical Applications”

- Humanoid robotics and robotic assistance
- Computer vision and multimodal robot perception
- AI and machine learning for robotic systems
- ROS 2 and robot control
- Integration and camera processing, LiDAR and other sensor data
- Development and evaluation of intelligent robotic assistance systems in medicine

August 2026 - Now

Magdeburg, Germany  
(Full-time)

### **Master Thesis – (University Clinic Magdeburg, Supervised by: Priv. Doz. Dr. Oliver Großer)**

“Optimization of Artifact Correction Parameters in Fluoroscopic CT Using Statistical Analysis and Decision Tree Modeling”

- Medical image processing
- DICOM analysis
- Data analysis and machine learning using R, RStudio and Python
- Statistical modeling
- Development of reproducible needle artifact analysis and correction workflow

Oct 2025 – July 2026

Magdeburg, Germany

Oct 2024 – Now

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| <b>Research Assistant – Medical Imaging &amp; Data Analysis (University Clinic Magdeburg, Supervised by: Priv. Doz. Dr. Oliver Großer)</b><br>“Medical imaging and data analysis lead by Siemens Healthcare” <ul style="list-style-type: none"> <li>• Fluoroscopic CT image processing using Siemens prototype software, Osirix and Image J</li> <li>• Data management</li> <li>• Data processing and analysis using R studio</li> <li>• Data analysis using Python 3</li> </ul>                                                                                                                                                                  | Magdeburg, Germany                                 |
| <b>Otto- von Guericke University of Magdeburg Research track project (Research Campus STIMULATE, Supervised by: Prof. Dr. Rose, 15 credits)</b><br>“The next generation operating room (OR) equipment management system: The smart aseptic barrier to enhance efficiency and patient care.” <ul style="list-style-type: none"> <li>• Development of an innovative and intelligent device for operating rooms that can minimize human mistakes</li> <li>• CAD modeling (SolidWorks) for the conceptual design</li> <li>• Hardware demonstrator creation</li> <li>• Programming a software for the device intelligence and functionality</li> </ul> | Oct 2023 – Oct 2024<br>Magdeburg, Germany          |
| <b>Mini research project (University Clinic Magdeburg, Supervised by: Priv. Doz. Dr. Oliver Größer, 5 credits)</b><br>“Evaluation of suitable contrast regulating substances for the creation of anthropomorphic phantoms and the context of possible training cases for interventional (CT)” <ul style="list-style-type: none"> <li>• In Lab Phantom creation and testing their standard Hounsfield Unit ranges</li> <li>• Collecting DICOM data sets from CT Images of the phantoms</li> <li>• Processing and analyzing the collected data using ImageJ and MATLAB</li> </ul>                                                                   | May 2023 – Oct 2023<br>Magdeburg, Germany          |
| <b>IGC - Medical engineer - Technical assistant in medical affairs</b> <ul style="list-style-type: none"> <li>• Report setting for international certificates on medical equipment manufacturers (QM Standards such as: ISO 9001, ISO 13485, and etc.)</li> <li>• Managing documentation and maintaining data integrity</li> <li>• Project planning</li> <li>• Work status tracking and deadlines using Jira</li> </ul>                                                                                                                                                                                                                           | Feb 2018 – Oct 2021<br>(Full-time)<br>Tehran, Iran |
| <b>Dey hospital – Medical engineering sector assistant</b> <ul style="list-style-type: none"> <li>• Repairing, maintenance and emergency interventions of all medical devices</li> <li>• Tracking device purchases and necessary upgrades</li> <li>• Maintaining contact with equipment supplies companies</li> <li>• Working with PACS and CMG M1 Pro</li> </ul>                                                                                                                                                                                                                                                                                 | Apr 2017 – Apr 2019<br>(Volunteer)<br>Tehran, Iran |
| <b>Jahan Gostaresh Tejarat - Technical assistant for device calibration</b> <ul style="list-style-type: none"> <li>• Calibrating medical devices such as: ECG, EEG, EMG and etc</li> <li>• Marketing for calibration services to the hospitals and keeping contact with customers for additional services and follow-ups</li> <li>• Tracking the work status aligning with ISO Standards</li> </ul>                                                                                                                                                                                                                                               | Apr 2016 – Apr 2017<br>(Part-time)<br>Tehran, Iran |

## SKILLS

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**Project Management:** MS Project, Trello

**Programming:** RStudio, MATLAB, Python

**Computer Aided Detection:** SolidWorks

**Medical Imaging:** MRI, CT-scan, Ultrasound, Fibroscan, EIT device, Dräger EIT tool

**Medical Image Processing:** ImageJ, Osirix, HOROS, Needle Interventions (Siemens)

**Data Processing and Visualization:** Power BI

**Medical Affairs:** ISO Standards (13485, 9001, etc.), QM Standards/ QM System (QMS) Software, Audit Management Software, FDA Standards, MDR Regulations

**Healthcare Technology:** PACs, HIS

**Design:** JavaScript, HTML, CSS

**Marketing:** Canva, Photoshop

**Others:** LATEX, Microsoft Office, Operating Systems (Linux, Mac, Windows)

**Languages:** English (C2), German (B2), Turkish (C1), Persian (Mother tongue)

## INTERESTS

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Cutting-edge Technology: AI, ML

Communication and Personal Development

Handling high-responsibility projects

Video and photo editing, Adobe Photoshop/ Premiere, Sports, Good food

## PATENT & PUBLICATIONS

**[1]** Robert Odenbach, Roland Croner, Mandana Kia, Calvin Yang | 2025 | *"Method of operating a medical surveillance system for handling and/or surveilling objects in a medical operation room and a surveillance system with enhanced functionality for application in a medical operation room and a corresponding apparatus"* | WO2025032258A1 | WIPO (PCT)  
<https://patents.google.com/patent/WO2025032258A1/en?q=WO2025032258A1>

**[2]** Journal of Electrical Bioimpedance | August 2019 Mohammad Karimi Moridani, Mandana Kia, and Fatemeh Choopani. *"Recognition of lung volume condition based on phase space mapping using electrical impedance tomography"* Journal of Electrical Bioimpedance 10.1 (2019): 34-39. DOI: <https://doi.org/10.2478/joeb-2019-0005>

**[3]** 3rd International Conference on Electrical Engineering | April 2019 Mohammad Karimi Moridani, Mandana Kia, and Fatemeh Choopani. *"Development of computer-aided diagnosis system to evaluate the lung efficiency using electrical impedance tomography."*